

CHEM 3610 (Rowley) Midterm Exam 3 Equations and Constants

Gas Constant Values			
8.314	$\frac{J}{mole\ K}$	0.08314	$\frac{L\ bar}{mole\ K}$
0.08206	$\frac{L\ atm}{mole\ K}$	8.314	$\frac{m^3\ Pa}{mole\ K}$
Boltzmann Constant Values			
1.381×10^{-23}	$\frac{J}{K}$	0.6950	$\frac{cm^{-1}}{K}$

Conversions		
1 L atm	=	101.325 J
1 atm	=	1.01325 bar
1 atm	=	760 torr
1 atm	=	101,325 Pa

$$\Delta G_{mix} = nRT (\chi_A \ln \chi_A + \chi_B \ln \chi_B)$$

$$\Delta S_{mix} = -nR (\chi_A \ln \chi_A + \chi_B \ln \chi_B)$$

$$p_A = \chi_A p_A^*$$

$$p_B = \chi_B K_B$$

$$\mu_A = \mu_A^* + RT \ln \chi_A$$

$$K_b = \frac{RT_b^{*2}}{\Delta H_{vap}}$$

$$\Delta T_b = K_b \chi_B = K_b C_B$$

$$K_f = \frac{RT_f^{*2}}{\Delta H_{freeze}}$$

$$\Delta T_f = K_f \chi_B = K_f C_B$$

$$\frac{n_g}{n_{total}} = \left| \frac{x_B - z_B}{x_B - y_B} \right|$$

$$\frac{n_l}{n_{total}} = \left| \frac{z_B - y_B}{x_B - y_B} \right|$$

$$\ln \gamma_A = \xi \chi_B^2 \quad \ln \gamma_B = \xi \chi_A^2$$

$$\mu_B = \mu_B^\ominus + RT \ln \gamma_B \chi_B$$

$$I = \frac{1}{2} \sum_{ions} z_i^2 \left(\frac{c_i}{c^\ominus} \right) \quad c^\ominus = 1\ molal$$

$$\log \gamma_{\pm} = -\frac{A\left| z_+z_- \right| \sqrt{I}}{1+B\sqrt{I}} + CI$$

$$\ln \frac{K_2}{K_1} = -\frac{\Delta H^\ominus}{R}\left(\frac{1}{T_2}-\frac{1}{T_1}\right)$$

$$\Delta G = \Delta G^\ominus + RT \ln Q$$

$$E_{\mathrm{cell}} = E_{\mathrm{cathode}} - E_{\mathrm{anode}}$$

$$\Delta G^\ominus = -RT \ln K_{eq}$$

$$\frac{\mathrm{d} E_{\mathrm{cell}}^\ominus}{\mathrm{d} T} = \frac{\Delta S^\ominus}{\nu F}$$

$$K_p=K_C\left(RT\right)^{\Delta n}$$

$$\nu_C E_C^\ominus = \nu_A E_A^\ominus + \nu_B E_B^\ominus$$

$$E_{cell} = E_{cell}^\ominus - \frac{RT}{\nu F} \ln Q$$

$$\Delta H^\ominus = -\nu F \left(E_{\mathrm{cell}}^\ominus - T \frac{\mathrm{d} E_{\mathrm{cell}}^\ominus}{\mathrm{d} T}\right)$$

$$J_{matter} = -D \frac{\mathrm{d} \mathcal{N}}{\mathrm{d} z} \qquad J_{energy} = -\kappa \frac{\mathrm{d} T}{\mathrm{d} z} \qquad J_{x\text{-momentum}} = -\eta \frac{\mathrm{d} v_x}{\mathrm{d} z}$$

$$D = \frac{1}{3} \lambda v_{mean} = \frac{1}{3} \left(\frac{k_B T}{\sigma p} \right) \left(\frac{8 R T}{\pi M} \right)^{1/2} \qquad \eta = \frac{1}{3} \lambda v_{mean} m \mathcal{N} = \frac{p M D}{R T}$$

$$\eta = \eta_0 e^{E_a/RT} \qquad \kappa = \frac{1}{3} \lambda v_{mean} \nu \mathcal{N} k_B = \frac{\nu p D}{T} \qquad \nu = \frac{1}{2} N_{D.o.F.}$$

$$G = \frac{1}{R} = \kappa \frac{A}{l} \qquad \Lambda_m = \frac{\kappa}{c} = (z_+ u_+ \nu_+ + z_- u_- \nu_-) F$$

$$f=6\pi\eta a \qquad u=\frac{ze}{f} \qquad s=uE=u\frac{\Delta V}{m}$$

$$c(x,t) = \frac{n_0}{A\sqrt{\pi Dt}} e^{-x^2/4Dt}$$

$$x_{rms} = \sqrt{2Dt}$$

$$F=96,485\frac{C}{mol} \qquad e=1.60217662\times10^{-19}C$$

[illegible]